



wergonic

SMART WORKWEAR SYSTEM

# User Manual

Ergonomic risk assessment & work  
technique training, powered by wearable  
motion sensors.

---

MANUAL

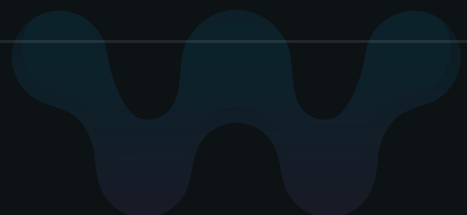
**Version 3.1**

MOBILE APP

**Android v5.1.7**

COVERS

**App & Web Dashboard**



## CONTENTS

# What's inside

---

|          |                                       |                                   |
|----------|---------------------------------------|-----------------------------------|
| <b>1</b> | <b>The Wergonic System</b>            | Overview & components             |
| <b>2</b> | <b>What's in the Kit</b>              | Hardware checklist                |
| <b>3</b> | <b>Sensors &amp; What It Measures</b> | IMUs, placement, metrics          |
| <b>4</b> | <b>System Setup</b>                   | App, pairing, phone, Work Library |
| <b>5</b> | <b>Running a Measurement</b>          | Fit, session, calibration, live   |
| <b>6</b> | <b>Reviewing Results</b>              | In the app & on the dashboard     |
| <b>7</b> | <b>Reference Values</b>               | Recommended limits & sources      |
| <b>8</b> | <b>Maintenance &amp; Return</b>       | Cleaning, storage, return         |
| <b>9</b> | <b>Troubleshooting &amp; FAQ</b>      | Common issues                     |

---



## INTRODUCTION

# The Wergonic System

Wergonic is a smart-workwear system for fast, accurate measurement of upper-body posture and motion during real work tasks — with results compared directly against recommended reference values.

### Smart Workwear

- Motion sensors fitted into garments for easy on/off use
- Tracks head, trunk, both arms, hand and forearm
- Estimates reachability (hand's reach)

### Mobile App · Android

- Streams and records sensor data over Bluetooth
- Real-time posture & movement metrics
- Annotate activities and breaks
- Syncs measurements to the web dashboard

### Web Dashboard

- Programme work cycles, activities and conditions
- Generate risk-assessment reports
- Built-in frameworks: RAMP, RULA, REBA, MEC (Volvo)
- AI-assisted interpretation of results
- User & organisation admin tools
- Runs in any browser — **cloud.wergonic.com**

**Your account.** You receive credentials to sign in to both the app and the dashboard with an email of your choice. Within an organisation, colleagues' accounts are linked for shared work.

## 2 HARDWARE What's in the Kit

The Wergonic Smart Workwear is delivered as a standardised kit.

- **Six IMUs** in colour-coded moulds
- **Android phone** with the Wergonic App installed and pre-paired
- **Wergonic shirts** in six sizes (XS–XXL)
- **Wergonic glove** and **headband**
- **Six CR2025** coin-cell batteries + one **phone charger**
- **Heart-rate chest band** with a mould-free IMU (optional)

### Phone compatibility

Use an Android phone on **Android 13 or newer**. Bluetooth hardware quality affects streaming. Tested models:

- Samsung Galaxy S21 FE · S21 Ultra
- Google Pixel 6A · Pixel 7A

**Compliance:** CE, FCC, IC · Bluetooth 4.0/5.0 · RoHS, REACH.



The six Wergonic / Movesense IMUs in their colour-coded moulds — Right arm Blue, Trunk Green, Head Black, Left arm Red, Hand Silver, Forearm Beige.

## Sensors & What It Measures

The system uses six Movesense inertial measurement units (IMUs), held in colour-coded moulds for correct placement.

### Sensor & specifications

- 9-axis motion (accelerometer, gyroscope, magnetometer) + temperature
- Accelerometer  $\pm 2/4/8/16$  g · Gyroscope  $\pm 125$ – $2000$  °/s
- Sampling 12.5–833 Hz · Bluetooth 4.0/5.0
- 36.6 mm  $\varnothing$  × 10.6 mm · 9.4 g · CR2025 battery
- Waterproof & shockproof

### Colour code

- Head — Black
- Trunk — Green
- Right arm — Blue
- Left arm — Red
- Hand — Silver
- Forearm — Beige

### 3.1 · Where each sensor goes

| BODY PART | COLOUR | PLACEMENT                                                                                     |
|-----------|--------|-----------------------------------------------------------------------------------------------|
| Head      | Black  | Centre of the <b>forehead</b> , held by the headband.                                         |
| Trunk     | Green  | Upper back, at the base of the neck (cervical/thoracic junction), in the shirt's back pocket. |
| Right arm | Blue   | Outer side of the right upper arm, over the deltoid, in the shirt's sleeve pocket.            |
| Left arm  | Red    | Outer side of the left upper arm, over the deltoid, in the shirt's sleeve pocket.             |
| Hand      | Silver | Back of the dominant hand, in the glove's hand pocket.                                        |
| Forearm   | Beige  | Dominant forearm, <b>just above the wrist</b> , in the glove's lower pocket.                  |

**Forearm and wrist are not the same thing.** There is no wrist sensor. The beige sensor is the **forearm** sensor — it sits on the forearm, close to the wrist, and that is what it is called everywhere in the app. The **wrist** is a result, not a sensor: it is calculated from the *Hand* and *Forearm* sensors together, as the difference in movement between them. That is why the report has a *Wrist* page even though no sensor is called Wrist.

### 3.2 · What is measured on each body part

All angles are inclinations measured against a **vertical reference** set during calibration. All speeds are **angular velocity** — how fast the body part moves, in degrees per second.

| BODY PART | ANGLES (°)                              | SPEED OF MOVEMENT (°/S) |
|-----------|-----------------------------------------|-------------------------|
| Head      | Forward/backward bending · Side bending | Not reported            |

|                            |                                                                                                                      |                                                                                                                      |
|----------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Trunk                      | Forward bending · Side bending                                                                                       | Not reported                                                                                                         |
| Right arm<br>Left arm      | Elevation of the upper arm (all shoulder movements, relative to vertical) · Forward bending · Side bending           | Arm angular velocity                                                                                                 |
| Forearm                    | Elevation of the forearm — used together with the upper-arm angle to estimate how far the hand reaches from the body | Feeds the wrist result below                                                                                         |
| Hand                       | —                                                                                                                    | Hand-transmitted vibration exposure                                                                                  |
| Wrist (calculated)         | Not reported as an angle                                                                                             | Wrist angular velocity — how fast the hand moves relative to the forearm, from the Hand and Forearm sensors together |
| Heart rate (optional band) | —                                                                                                                    | Heart rate, and time spent above 33% of heart-rate reserve                                                           |

**Rotation is not measured.** The system does not measure axial rotation — twisting of the trunk, turning of the head, or rotation of the upper arm. Every angle in the report is an *inclination* against vertical. The only rotational figure is the wrist's angular velocity, which is the speed of the hand turning relative to the forearm, not a twist angle.

**Live tiles vs report.** During recording, head, trunk and arm *side bending* and arm *forward bending* are shown as extra tiles — switch them on with **Show extra tiles** in Settings. The full report is covered in chapter 6, and the recommended limits for each quantity in chapter 7.

# 4

## GETTING STARTED

# System Setup

Use the provided phone (mostly pre-configured) or your own Android phone. Either way, finish by setting up your Work Library on the dashboard.

### 4.1 · Using the provided phone

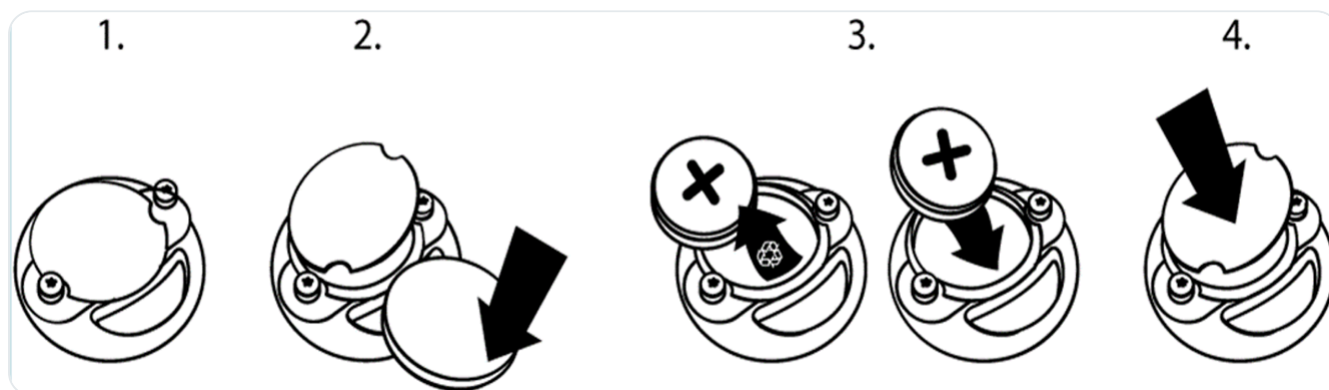
The kit phone already has the correct Android version, the latest Wergonic App and pre-paired IMUs. Two steps remain:

1. **Sign in to Google Play** with a Gmail account so the app stays updated with new releases.
2. **Set up your Work Library** on the dashboard (see 4.4).

Resetting the phone to factory settings is optional. If you do, follow the "own phone" steps below afterwards.

### 4.2 · Using your own phone

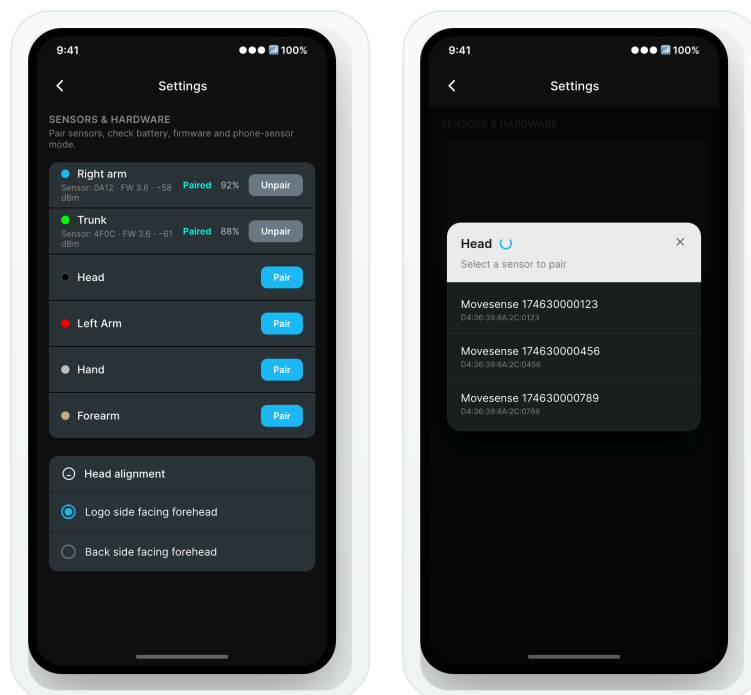
1. **Install the app** — open Google Play, sign in, search **Wergonic**, install the official app.
2. **Start the IMUs** — insert a CR2025 into each unit (pry the lid with a coin). A **red light blinks** when the IMU is on; it can take up to 30 seconds to appear.
3. **Sign in** to the app with your credentials.



Opening the battery lid and inserting a CR2025 coin cell — use a coin (or small spoon) to pry the lid, then reattach it.

### 4.3 · Pairing the IMUs

Open **Settings** (gear icon, top-right of the home screen) and go to **Sensors & Hardware**. Each body part has a **Pair** button; tap it and pick the sensor from the list. Identify a sensor by the last four digits of the serial number on its back. Once connected, the row shows **Paired** in cyan with battery level and an **Unpair** option.



**A** Settings → Sensors & Hardware. Pair / Unpair each body part; "Head alignment" sits below.

**B** "Select a sensor to pair" — tap the matching serial number to connect it to that body part.

**Colour-code standard** — Head Black · Trunk Green · Right arm Blue · Left arm Red · Hand Silver · Forearm Beige. The optional HR chest band pairs with the mould-free sensor.

**Head alignment.** The head IMU can face either way on the headband. In Settings, set **Logo side facing forehead** (logo turned inwards, towards the skin) or **Back side facing forehead** (logo turned outwards) to match how you fitted it.

**Test auto-pairing.** Log out, close and reopen the app — sensors should reconnect automatically. If you're not measuring right away, remove the batteries to save charge.

### Phone settings (own phone)

- **Battery:** turn off adaptive/optimised modes; set the Wergonic App to **Unrestricted**.
- **Permissions:** Location and Nearby devices → **Allow all the time**.
- **Notifications:** enable for the Wergonic App.
- **Unused-apps / app-pausing:** disable for the Wergonic App.

## 4.4 · Set up your Work Library on the dashboard

Sign in at [cloud.wergonic.com](https://cloud.wergonic.com). In the left menu, under **Setup**, open **Work Library**. This is where you prepare everything the app offers you during a measurement. The page has three tabs:

| TAB         | WHAT IT HOLDS                                                                                                                                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Work Cycles | Your factories and the work they contain, as a tree: <b>Factory</b> → <b>Line</b> → <b>Workstation</b> → <b>Operator</b> → <b>Work cycle</b> → <b>Tasks</b> . Each task has a name, an optional description and a duration in seconds. <b>Required</b> when you measure a work cycle. |



|                   |                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Activities</b> | Activity groups and the activities inside them. These are the buttons you tap during a free recording to mark what the worker is doing. Optional, and only available when the phone is online. |
| <b>Conditions</b> | Condition groups and their values — the circumstances you want to tag a measurement with (single-choice or multiple-choice). Optional.                                                         |

**Adding work cycles.** On the *Work Cycles* tab use **Add factory** to build the tree by hand, then add lines, workstations, operators, work cycles and tasks step by step. To load many at once, prepare them in Excel and use **Import CSV/Excel**.

**Adding activities and conditions.** On the *Activities* or *Conditions* tab, first create a group, then add items to it. Items can later be renamed, merged into another item, or retired.

**Menu names.** The left menu is grouped into **Overview** (Home), **Measurements**, **Risk assessments**, **Workers**, **Setup** (Work Library) and — for organisation admins — **Team**. This manual uses those exact names.

## Running a Measurement

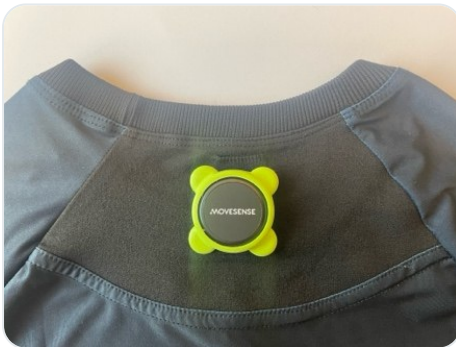
### 5.1 • Prepare

- Fully charged phone and six paired IMUs
- Fresh CR2025 batteries (recommended)
- Clean Wergonic shirt & glove in the right size, plus a headband for the head IMU
- A ~2 kg weight for arm calibration

### 5.2 • Start the IMUs & fit the workwear

Insert batteries and confirm each IMU shows a **red blinking light**. Unless stated otherwise, fit each sensor with the Movesense logo **facing outward and upright**:

- **Head (black) — headband.** Put the headband on so the sensor sits in the **centre of the forehead**, just above the eyebrows and below the hairline. *Not* over the ear and not on the back of the head. The sensor may face either way; whichever you choose, set **Head alignment** in Settings to match (see 4.3).
- **Trunk (green) — shirt.** Slide the sensor into the pocket on the **upper back**, at the base of the neck. The shirt should fit snugly so the sensor cannot slide.
- **Arms (blue, red) — shirt sleeves.** Right arm blue, left arm red, in the pocket on the outer side of each upper arm.
- **Hand (silver) and Forearm (beige) — glove.** Silver on the back of the hand, beige in the lower pocket on the forearm just above the wrist. Both are worn on the **dominant** arm.

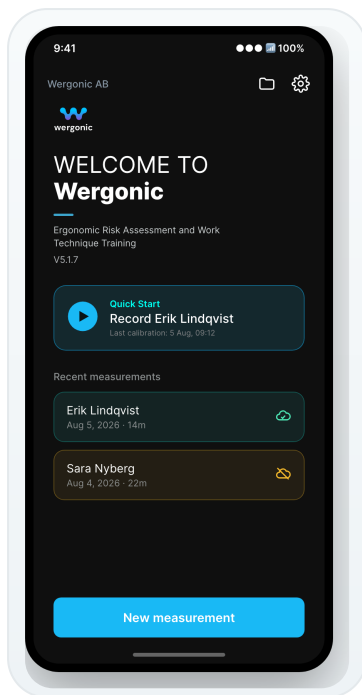


Trunk IMU seated in the shirt's upper-back pocket at the base of the neck — logo facing outward and upright.

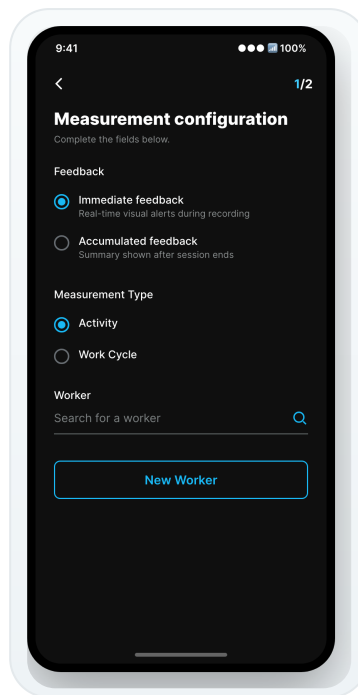
**The hand and forearm sensors work as a pair.** Connect both, or neither. The wrist result is calculated from the two together — with only one of them connected there is no wrist result at all.

### 5.3 • Start a new measurement in the app

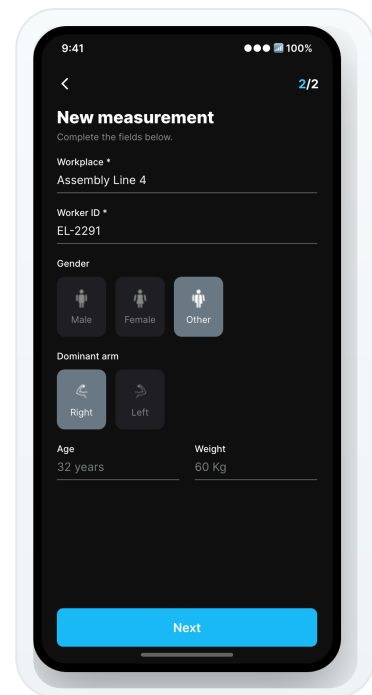
On the home screen tap **New measurement**, then complete the two setup pages and confirm pairing before calibration.



**C** Home — tap **New measurement**. The folder icon opens history; the gear opens settings.



**D** Page 1 — two separate choices: **Feedback** (how the worker is warned while recording) and **Measurement Type** (how the work is logged). Then pick or search a worker.



**E** Page 2 — worker profile (workplace, ID, gender, dominant arm, age, weight). Tap **Next**.

## Page 1 in detail — Feedback and Measurement Type are independent

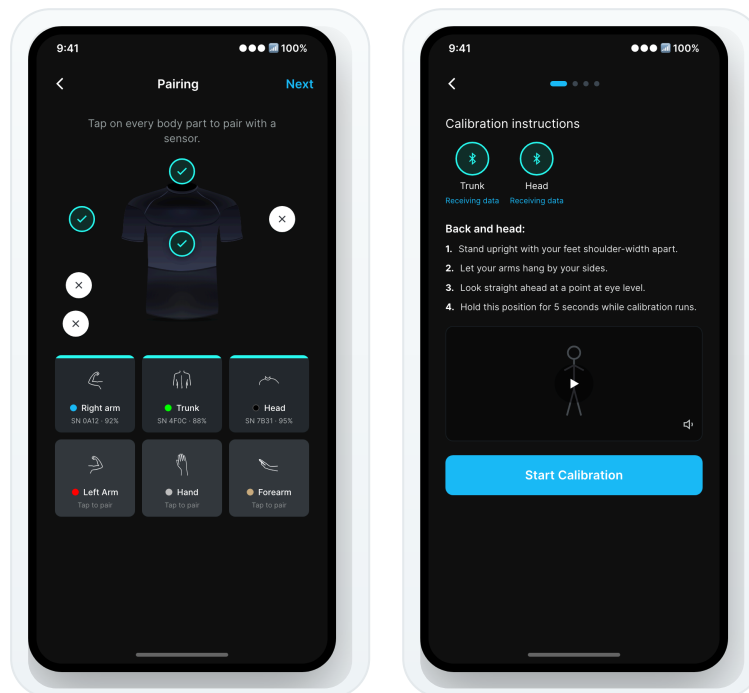
They answer two different questions. You set both, in any combination.

| SETTING                                                                 | OPTION                      | WHAT IT DOES                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Feedback</b><br>How the worker is warned <i>during the recording</i> | <b>Immediate feedback</b>   | Real-time alerts the moment a limit is passed — the metric square turns orange or red, and vibration or voice cues fire if you enabled them in Settings. Use this for work-technique training.                                                                   |
|                                                                         | <b>Accumulated feedback</b> | No alerts while working. The worker is left undisturbed and gets a summary once the measurement ends. Use this when you want to record normal work without influencing it.                                                                                       |
| <b>Measurement Type</b><br>How the work is logged                       | <b>Activity</b>             | Record freely and log activities as they happen. Nothing is planned in advance — you tap the activity being done now, and the previous one ends automatically.                                                                                                   |
|                                                                         | <b>Work Cycle</b>           | Follow a predefined task list for a station. Pick factory, line, workstation, operator and model first, then step through the tasks manually or let the app advance them automatically. Requires a work cycle prepared in the Work Library, and an online phone. |

**Online vs offline.** To load a pre-programmed **Work Cycle** or reuse a saved worker profile you must be online. Offline, create a new profile and sync later.

## 5.4 • Confirm pairing, then calibrate

The **Pairing** screen shows the shirt with each body part. Paired sensors show a **cyan check**; tap a body part to pair, or tap a paired one to unpair. When ready, tap **Next** (top-right) to calibrate.



**F** Pairing overview — connection cards show each sensor's serial and battery. Tap **Next** when the sensors you need are connected.

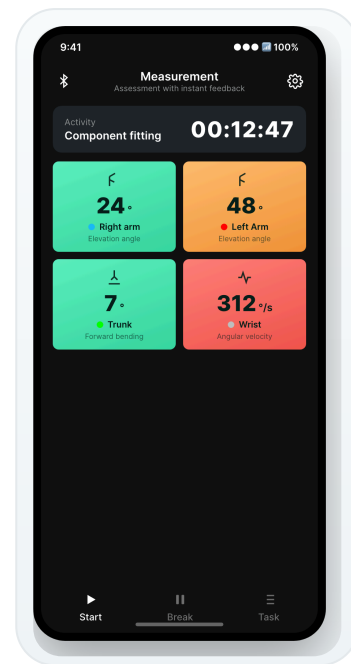
**G** Calibration — follow the numbered steps and demo video, then **Start Calibration** and hold the pose for 5 seconds.

Calibration is done per test person, in **four steps** — **Back and head**, **Right arm**, **Left arm**, and a **Forward reference**. Each step gives text, a demo video (with audio) and voice cues; the arm steps use the ~2 kg weight. Calibration is saved on the phone and can be reused for consecutive measurements on the same person **if the IMUs have not moved** — otherwise recalibrate.

## 5.5 • Live mode & recording

After calibration you enter live mode. Metric squares update in real time and colour by risk — **green** (safe), **orange**, **red**.

- **Choose metrics:** long-press any square, tick the metrics you want, then **Save** (top-left).
- **Recalibrate:** before starting (timer at 00:00:00), tap the gear → a body part.
- **Record:** tap **Start**; use **Break** to pause; tap **Finish** to end and save.
- When online you can also add activities during recording.



**H** Live mode — Start / Break / Finish controls, with risk-coloured metric squares.

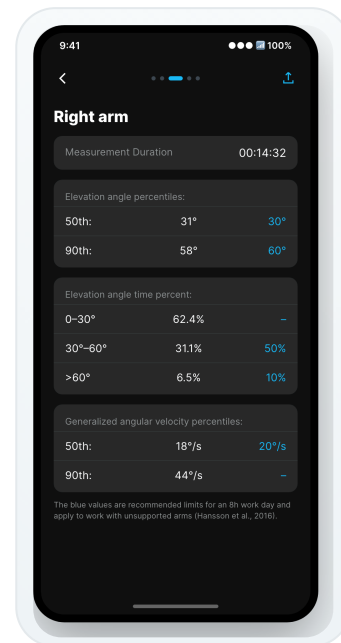
## Reviewing Results

Review a measurement directly in the app, or in full on the web dashboard.

### In the app

Tap the **folder icon** (home, top-right) to open your measurements. On a measurement, tap the **: menu** → **Report**. The report opens as **swipeable pages** — one per body part — with the numbers in tables. Recommended limits are shown in **blue** next to your result; all of them are listed in chapter 7.

- The **: menu** also offers **Sync measurement**, **Share** and **Delete**.
- Tapping a measurement opens its **timeline** (activities, breaks).



**i** In-app report — per-body-part tables; blue values are the recommended 8-hour limits.

### On the web dashboard

Online measurements upload automatically; offline ones upload when you reconnect (may take a few minutes). Sign in at **cloud.wergonic.com** and open **Measurements** in the left menu. The table lists every measurement with its worker, recorded body parts, duration, start time, risk assessments and status. **Click a finished measurement** to open its details.

Results are shown as:

- **Pie charts** — posture (°) first; switch to *Speed of movement* for angular velocity (°/s). Export to Excel/PDF, or open *Show Detailed View* for time plots.
- **Tables** — posture, an algorithm-based reach estimate, then speed of movement.

The row menu on each measurement also offers **Run risk assessment**, **Generate report**, **Export raw data** and **Archive**. Select several measurements to **compare** them or build a **combined report**.

## Reference Values

These are the recommended limits the app shows in blue beside your results. They apply to a full **8-hour working day**. Exceeding one is not a diagnosis — it is a signal that the exposure should be reduced.

### 7.1 • How to read the numbers

- **50th percentile (median)** — the value the worker was at or below for half of the measured time. This is the typical working posture or speed.
- **90th percentile** — the value reached during the most demanding 10% of the time. This captures the peaks.
- **Time percent** — how much of the measured time was spent inside each angle band.

### 7.2 • Recommended limits

| BODY PART                  | QUANTITY                                               | RECOMMENDED LIMIT                                                                                    | SOURCE                 |
|----------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------|
| Arms<br>(unsupported)      | Elevation angle — 50th percentile                      | $\leq 30^\circ$                                                                                      | Hansson et al., 2016   |
|                            | Elevation angle — 90th percentile                      | $\leq 60^\circ$                                                                                      |                        |
|                            | Angular velocity — 50th percentile                     | $\leq 30^\circ/\text{s}$                                                                             | Arvidsson et al., 2017 |
| Trunk                      | Forward bending — 50th percentile                      | $\leq 20^\circ$                                                                                      | Hansson et al., 2016   |
|                            | Side bending — 50th percentile                         | $\leq 20^\circ$                                                                                      |                        |
| Head                       | Side bending — 50th percentile                         | $\leq 20^\circ$                                                                                      | Hansson et al., 2016   |
| Wrist                      | Angular velocity — 50th percentile                     | $\leq 20^\circ/\text{s}$<br>$\leq 15^\circ/\text{s}$ if the work is also assessed as force-demanding | Arvidsson et al., 2017 |
| Heart rate (optional band) | Accumulated time above 33% of heart-rate reserve (HRR) | Keep as low as possible                                                                              | —                      |

### 7.3 • Angle bands used in the report

The report also shows how much of the time was spent in each band. The lowest band starts at the calibrated upright posture and includes a small backward margin, so a slight lean backwards is still counted as neutral.

| QUANTITY                | BANDS REPORTED                                                               |
|-------------------------|------------------------------------------------------------------------------|
| Trunk — forward bending | $10^\circ\text{--}20^\circ \cdot 20^\circ\text{--}45^\circ \cdot > 45^\circ$ |

|                        |                           |
|------------------------|---------------------------|
| Trunk — side bending   | 0°–10° · 10°–30° · > 30°  |
| Head — forward bending | 10°–20° · 20°–30° · > 30° |
| Arm — elevation        | 0°–30° · 30°–60° · > 60°  |

Trunk bands follow the thresholds used in the RAMP tool (Lind et al., 2020).

## 7.4 • Full references

These are the sources cited in short form inside the mobile app.

1. Arvidsson, I., Dahlqvist, C., Enquist, H., & Nordander, C. (2017). *Åtgärdsnivåer mot belastningsskada* [Action levels against work-related musculoskeletal disorders]. Arbets- och miljömedicin Syd, Report No. 18/2017. Lund, Sweden.
2. Hansson, G.-Å., Arvidsson, I., & Nordander, C. (2016). *Riktvärden för att bedöma risken för belastningsskador, baserade på tekniska mätningar av exponeringen* [Guideline values for assessing the risk of work-related musculoskeletal disorders, based on technical measurements of the exposure]. Arbets- och miljömedicin, Lund, Report No. 4/2016.
3. Lind, C. M., Forsman, M., & Rose, L. M. (2020). Development and evaluation of RAMP II — a practitioner's tool for assessing musculoskeletal disorder risk factors in industrial manual handling. *Ergonomics*, 63(4), 477–504.





CARE

## Maintenance & Return

### Between sessions

- **Remove the IMUs**, then machine-wash garments at **30 °C** with regular, perfume-free detergent (no disinfecting softener).
- **Disinfect** the removed IMUs.
- Avoid fully discharging the phone.

### Returning the kit

- Clean the workwear as above.
- Batteries needn't be returned — recycle any you won't reuse.
- Fully charge the phone.
- Log out of Google Play and the Wergonic App.



# Troubleshooting & FAQ

## Troubleshooting

| PROBLEM                             | WHAT TO DO                                                                                                                                                    |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IMU won't blink / seems off         | Moist fingers can discharge coin cells. Try a fresh battery; if it still won't start, contact Wergonic for a replacement.                                     |
| IMU is slow to pair                 | Start the IMU <i>before</i> opening the app and wait ~1 minute. Don't repeatedly tap pair/unpair. Try another location — some environments disturb Bluetooth. |
| Battery lid won't open              | Use only CR2025, inserted the right way up. Never use CR2032 — it can jam and damage the lid.                                                                 |
| App stuck loading after calibration | Usually an offline network search. Enable flight mode <i>with Bluetooth on</i> , or work online for best stability.                                           |
| No wrist result in the report       | The Hand and Forearm sensors work as a pair. If only one of them was connected, no wrist value can be calculated.                                             |

## Frequently asked questions

### Do I need to be online to measure?

No — only Bluetooth is required. Data is stored on the phone and uploads when you're back online. Online mode is more stable and is recommended.

### How far does Bluetooth reach?

Roughly 10 metres, but range depends on the environment where you measure.

### Does the system measure rotation?

No. It measures inclination angles against vertical and speed of movement. Twisting of the trunk, head or upper arm is not measured — see chapter 3.2.

### Why does the app say "Forearm" but the report says "Wrist"?

Because they are two different things. The beige sensor sits on the forearm and is called Forearm everywhere it is set up. There is no wrist sensor — the wrist figure is calculated from the Hand and Forearm sensors together. See chapter 3.1.

### Where exactly does the head sensor go?

In the centre of the forehead, held by the headband — not over the ear. See chapter 5.2.

**Need help?** Contact the Wergonic team for hardware replacements or support with setup and measurements.